Vision Through Technology: 'Digi Drishti's' Pioneering Role in Assisting the Visually Impaired

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***Abstract*— The "Digi Drishti" project is a groundbreaking initiative in assistive technology designed to assist visually impaired individuals in public transportation systems. The project aims to bridge the gap between advanced technology and the practical needs of this demographic, enhancing their independence and quality of life. The core of the project is its innovative use of image processing and machine learning algorithms for currency recognition, which uses a high-resolution camera and a Convolutional Neural Network (CNN) to accurately identify currency denominations. The device also integrates QR code scanning capabilities, providing real-time navigation assistance in public transportation. The design philosophy is user-centric, focusing on intuitive operation, reliability, and accessibility. The hardware architecture includes a Raspberry Pi microcontroller, a dual-camera module, and an audio output system, while the software architecture is a blend of operating systems, programming languages, and libraries. The development of "Digi Drishti" followed a structured and iterative approach, including conceptualization, design, implementation, and rigorous testing. The results have been overwhelmingly positive, with high accuracy rates in currency recognition and QR code scanning, even under challenging conditions. However, the project identified areas for improvement, such as enhancing performance in low-light conditions and refining the ergonomic design. "Digi Drishti" demonstrates the potential of technology to make a tangible difference in the lives of visually impaired individuals and demonstrates how innovative solutions can be developed by understanding and addressing specific challenges.**

Keywords*— *Assistive Technology, Visual Impairment, Currency Recognition, QR Code Navigation, Convolutional Neural Networks (CNN), Image Processing, User-Centric Design, Public Transportation Accessibility.

I. Introduction

The "Digi Drishti" project aims to address the challenges faced by the visually impaired, a population of approximately 285 million people worldwide, with 39 million being completely blind. The development of assistive devices using advanced technology in image processing, machine learning, and mobile computing can significantly improve the autonomy and quality of life for this population. Two primary challenges impacting the independence of visually impaired individuals are currency recognition and navigation in public transportation systems. Existing solutions often lack holistic solutions, particularly in terms of real-time processing, accuracy, and user-friendly interfaces.

The proposed system aims to design and implement a robust image processing algorithm with a minimum of 95% accuracy in currency note recognition, integrate efficient QR code scanning capabilities for real-time public transportation navigation, develop a portable, ergonomic hardware design incorporating a Raspberry Pi, a high-resolution camera, and an intuitive user interface, and ensure the software architecture is scalable, maintainable, and easily upgradable. The report covers the project's theoretical underpinnings in computer vision and machine learning, practical considerations in hardware integration and user interface design, a thorough literature review, detailed system architecture, development methodology, and analytical testing and validation. It concludes with a critical evaluation of the project outcomes, challenges encountered, future research avenues, and potential societal impact of the device.

Fig. 1. Currency Identification System for the Visually Impaired

The above Figure 1 depicts how the device will work to tell the denomination of a banknote to the user, and also to tell if it's a fake or real note. The device captures the image of the note using its Camera and gives Audio Output of its value.

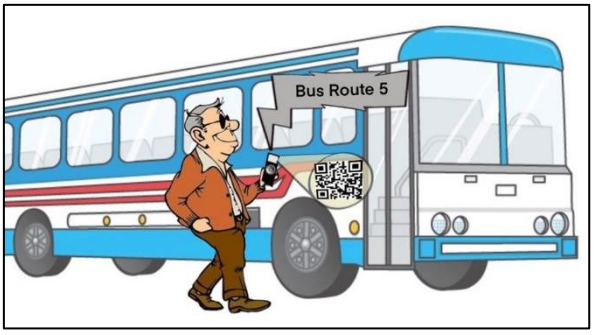

Fig. 2. Transport Information System for the Visually Impaired

Figure 2 shows how the device can be used to tell public transport route and schedules. A QR Code is placed at the entry of a transport vehicle. When the visually impaired person uses the device to scan the QR Code before getting on the vehicle, the essential Transport Information is given out as Audio.

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II. LITERATURE REVIEW

This literature review provides an in-depth analysis of current assistive technologies, focusing on their technical composition, operational efficacy, and user experience. It discusses various technologies such as Optical Character Recognition (OCR) algorithms, Convolutional Neural Networks (CNNs), GPS-based systems, RFID and ultrasonic sensors, emerging technologies like LiDAR, and user interface design principles. The review also explores the limitations of current image recognition systems in handling diverse environmental conditions, emphasizing the need for adaptive algorithms that can dynamically adjust to different lighting, note conditions, and even handle multiple currencies. The review also evaluates real-time processing and latency issues in existing navigation systems, stressing the importance of real-time data processing, especially in dynamic environments like public transportation.

The review highlights the disparity between the availability of advanced technologies and their affordability, discussing the market gap for cost-effective solutions that provide state-of-the-art features, especially in low-resource settings. It also addresses the challenges in integrating various technologies into a cohesive system, emphasizing the importance of interoperability and modular design in creating versatile and scalable assistive devices. The theoretical framework for the development of "Digi Drishti" is outlined, which includes advanced image processing techniques, human-computer interaction (HCI) principles, mobile computing and its evolution, and sociotechnical systems in assistive technology. Advanced image processing techniques, particularly deep learning techniques like CNNs, are discussed for their efficacy in complex pattern recognition tasks, particularly in accurate currency recognition. Human-computer interaction (HCI) principles are explored, including user-centered design, multimodal interaction, and adaptive interfaces.

Vision impairment is a leading cause of disability worldwide. India is now home to the world's largest number of blind people. Of the 37 million people across the globe who are blind, over 15 million are from India. The global market size and growth for Assistive Technologies for Visually Impaired industry is anticipated to surge at a remarkable rate. A survey conducted by NCBI showed that more than 15% of the poeple facing blindness of different forms in India are above the age of 60 who require assistance. The projected number of people with vision loss is also said to increase drastically by the year 2050 of around 1.5 billions in number.

Meharu et. al. [2] aims to develop a Convolutional Neural Network (CNN) model to identify Ethiopian banknotes, particularly for real-time recognition. The model is designed to recognize currency notes even in partially distorted, folded, or challenging conditions. A dataset of 8500 Ethiopian banknote images was collected with the help of 9 blind individuals. The model was trained and evaluated using TensorFlow Object Detection API and pre-trained models such as Faster R-CNN Inception and SSD MobileNet. The entire codebase was implemented in Python. The models were tested with 500 real-time video sequences under different banknote quality and lighting conditions. The Faster R-CNN Inception model achieved an average accuracy of 91.8% in real-time video testing. This study represents a significant step towards empowering the visually impaired in Ethiopia with real-time currency recognition capabilities.

V, C, et. al. [1] discusses the use of camera-based computer vision technology to help visually impaired individuals identify paper currency. It identifies the need for a precision detection algorithm and adaptability to various environments. Current algorithms are limited to specific environments and have limitations. The proposed solution is a deep learning approach with improved data collection for class-specific features, enhanced handling of occlusions and changes in perspective, and demonstrated efficiency in dealing with rotations, scaling, and illumination variations. The paper presents a new method for recognizing Indian currency notes, utilizing deep learning and transfer learning techniques for feature extraction. The project structure involves training the model using a pre-trained architecture and using a camera to capture the currency image. The system successfully recognizes paper currency.

Joshi, R.C. et. al. [3] proposed solution to the challenge of recognizing banknotes is a YOLO-v3 CNN model-based system. This system collects and processes images of banknotes in various denominations and conditions, enhancing its robustness. The model is then trained and evaluated in real-time scenarios and on a test dataset. The system achieved a detection accuracy of 95.71% and recognition accuracy of 100%, making it a standalone, real-time solution for banknote identification. This system is crucial for visually impaired or blind individuals to conduct business transactions independently.

III. PROJECT DESCRIPTION

A. *Design Philosophy*

"Digi Drishti" is a user-centric project aimed at bridging the gap between advanced technology and practical usability for visually impaired individuals. Its key aspects include inclusivity and accessibility, reliability and accuracy, portability and durability, and scalability and upgradability. The device is designed to be intuitive, easy to use, and reliable, with robust algorithms and sensor systems that perform consistently under various environmental conditions. It is lightweight, compact, and durable, making it suitable for daily use and easy to carry. The architecture is designed to be scalable, allowing for future enhancements and integration with additional functionalities or technologies.

B. *System Architecture*

Digi Drishti is an assistive device that combines hardware and software components to create a cohesive and efficient system. The hardware includes a microcontroller, a camera module, an audio output system, additional sensors like GPS and gyroscopes, and a power supply. The software architecture includes an operating system, currency recognition software, navigation software, and user interface software. The operating system is based on factors like compatibility, performance, and support for libraries. The currency recognition software focuses on image processing and machine learning algorithms for currency detection. The navigation software handles GPS data and QR code scanning, while the user interface software manages user inputs and provides audio feedback.

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C. Component

The "Digi Drishti" project incorporates a range of components, each playing a vital role in the system's overall functionality. These components are carefully selected to ensure optimal performance, reliability, and user experience. The key components include:

- Raspberry Pi Microcontroller: Serves as the central processing unit of the system, handling computations, data processing, and controlling other hardware components.
- Camera Module: A critical component for the image capture necessary for currency recognition and QR code scanning. It captures high-resolution images which are then processed by the software.
- Audio Output System: Essential for providing auditory feedback to the user, including currency denomination information and navigation instructions.
- GPS Module: Facilitates location tracking and navigation assistance, crucial for the real-time public transportation information feature.
- Battery: Powers the device, ensuring that it remains operational for extended periods, suitable for daily use.
- Software: Encompasses the operating system and various software applications and libraries. This includes image processing software for currency recognition, QR code scanning algorithms, and user interface software for interaction with the device.

Each of these components contributes to the "Digi Drishti" system, working in unison to provide a seamless and intuitive experience for visually impaired users.

IV. Methodology

A. Development Process

Digi Drishti, a system designed for visually impaired individuals, was developed using a structured and systematic approach. The project began with an analysis of the needs of visually impaired individuals, involving consultations with experts. A comprehensive requirement analysis was conducted to define the system's functional and non-functional requirements, including currency recognition accuracy, real-time navigation processing, user interface design, portability, and power efficiency. The design phase involved creating detailed blueprints of the system architecture, including the Raspberry Pi microcontroller, a high-resolution camera, and a GPS module for navigation. The software design included planning the operating system, development environment, and image processing and navigation software architecture. Early prototypes were developed to test the feasibility of the hardware and software components, which underwent iterative testing and refinement. The implementation phase involved the coding of the software, including machine learning algorithms for currency recognition and QR code scanning and navigation. The system underwent rigorous testing, including functional testing, integration testing, and user testing with visually impaired individuals, to refine the system further.

B. Software Architecture

Digi Drishti is a robust, scalable, and user-friendly software architecture designed for Raspberry Pi. The project's operating system was chosen to be lightweight and efficient, with a Linux distribution for compatibility and support for various programming languages and libraries. The development environment was equipped with necessary tools and libraries, including machine learning frameworks for currency recognition. The currency recognition software involved developing a machine learning model trained on a dataset of currency notes, responsible for image preprocessing, feature extraction, and classification. The navigation software processed GPS data and decoded QR codes, offering real-time location tracking, route mapping, and audio feedback. The user interface software focused on simplicity and ease of use, including voice commands and tactile buttons for visually impaired users. The software modules were integrated and tested extensively to ensure seamless operation, including interaction between software and hardware components and overall performance.

C. Hardware Integration

The "Digi Drishti" system was developed by integrating hardware components. The assembly process involved integrating the Raspberry Pi microcontroller with a camera module, GPS module, and audio output system. The circuit design was crucial for efficient operation, involving layout design, component selection, and prototyping. Power management was also a key consideration, ensuring a power supply system that could manage component power consumption and ensure a long battery life. The hardware-software integration was the final step, ensuring the software could effectively control hardware components and process sensor data. This integration was crucial for the system's functionality and reliability.

V. Currency Recognition

The Digi Drishti system's Currency Recognition module uses image processing and deep learning to accurately identify currency denominations. The module uses a high-resolution camera for detailed image capture, with techniques like Histogram Equalization, Contrast Enhancement, and Noise Reduction for improved clarity. Feature extraction includes pattern recognition, edge detection, contour analysis, and segmentation[9]. A Convolutional Neural Network (CNN) is trained on a diverse dataset of currency note images to ensure robust learning[4][5]. The CNN classifies the notes based on their learned features, computing probability scores for each denomination and selecting the highest score as the identified denomination. User feedback is provided through audio output using Text-to-Speech (TTS) technology, informing the user of the currency denomination. If the classification confidence is below a certain threshold, the system advises the user to adjust the note's position or orientation and retry[7].

The module is integrated with the main system, ensuring compatibility and seamless interaction with other functionalities[8]. Accuracy testing is conducted across various note conditions and lighting scenarios, with a target accuracy rate of above 95%. Performance testing optimizes the processing time from image capture to audio feedback, with a target time under 2 seconds for a seamless user experience[6].

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i. **Start**: The currency recognition process begins.

ii. **Image Capture**:

- **Input**: The camera captures an image of the currency note.
- **Action**: The system activates the camera module to take a high-resolution image.

iii. **Preprocess Image**:

- **Action**: The captured image is normalized and enhanced.
- **Steps**: a. Apply Histogram Equalization for normalization. b. Use CLAHE for contrast enhancement. c. Implement Gaussian Blur for noise reduction.

iv. **Feature Extraction**:

- **Action**: Extract relevant features from the preprocessed image.
- **Steps**: a. Apply Canny Edge Detection to identify edges. b. Perform Contour Detection to find the contours of the currency note.

v. **Classify Currency**:

- **Input**: Extracted features from the image.
- **Action**: Use a trained CNN model to classify the currency denomination.
- **Process**: The CNN model analyzes the features and classifies the image into a currency denomination based on learned patterns.

vi. **Provide User Feedback**:

- **Action**: Communicate the classification result to the user.
- **Steps**: a. If the currency is recognized, announce the denomination through audio output. b. If recognition fails, prompt the user to adjust the note or retry.

vii. **End**: The currency recognition process concludes.

Fig. 3. Algorithm for Module 1: Currency Recognition

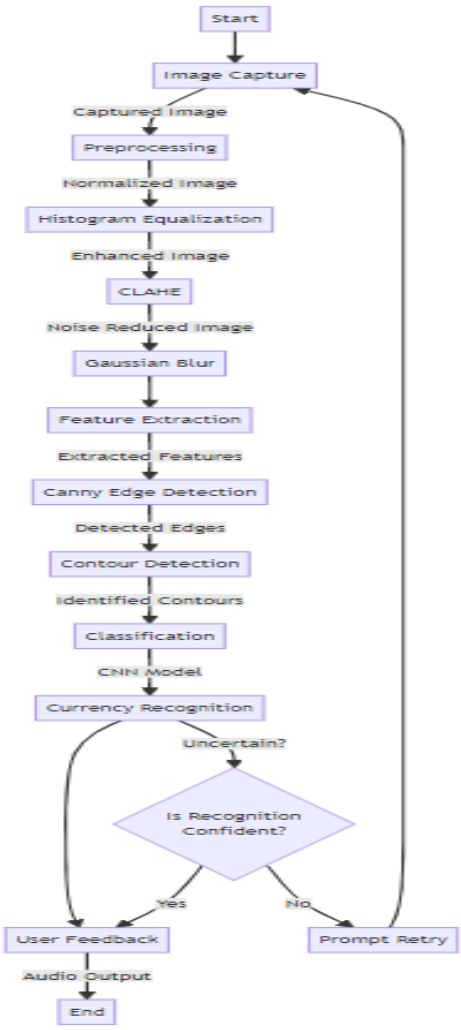

Fig. 4. FlowChart for Module 1: Currency Recognition

The Digi Drishti project's Currency Recognition module is a sophisticated system designed to assist visually impaired individuals in identifying currency denominations. The process begins with an image capture stage, where a high-resolution camera captures an image of the currency note. The image undergoes preprocessing steps, including normalization, contrast enhancement, and noise reduction using Gaussian Blur. The system then identifies and isolates key features of the currency note using the Canny Edge Detection algorithm and contour detection algorithms.

The classification process involves a Convolutional Neural Network (CNN) trained on a diverse dataset of currency notes. The CNN model uses its learned patterns and features to determine the most likely denomination of the currency note. The result of this classification is communicated to the user through an audio output, ensuring user-friendly interaction and accuracy. This module exemplifies the integration of advanced image processing and machine learning techniques to enhance the financial independence of visually impaired users.

The currency recognition module starts with an image capture using a high-resolution camera. The image is ormalized using Histogram Equalization, Contrast Limited Adaptive Histogram Equalization (CLAHE), and Gaussian Blur to enhance contrast. The preprocessed image is then subjected to Canny Edge Detection and Contour Detection to identify unique features

The extracted features are then classified into a Convolutional Neural Network (CNN) model, which classifies the note into a specific denomination based on the identified features. The system then provides audio feedback to the user if the recognition is confident, or enters a decision loop if it is uncertain. If the recognition is confident, the system provides feedback, prompts the user to retry, and the process concludes after providing feedback or prompting a retry. This process ensures accurate and reliable recognition of currency notes.

VI. MODULE 2 REAL-TIME NAVIGATION ASSISTANCE IN PUBLIC TRANSPORTATION

Module 2 is a navigation system designed to assist visually impaired users in public transportation environments. It uses GPS technology and QR code scanning to provide accurate and timely information about bus routes, schedules, and locations. The system tracks the user's real-time location, continuously processing data to determine their proximity to public transportation stops. QR code images are captured and decoded to extract route information, including bus numbers, routes, and schedules[10]. The system retrieves real-time data from transportation databases or APIs, providing up-to-date information about bus schedules, delays, and other relevant updates. The retrieved data is processed and correlated with the user's location and decoded QR code information to provide relevant navigation assistance[11][12]. The system's

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user interface is both audio and interactive, allowing users to request specific information or confirm their current location. It is seamlessly integrated with the core "Digi Drishti" system, ensuring compatibility with other functionalities like currency recognition. The module undergoes extensive testing to ensure accuracy in GPS tracking, efficiency in QR code scanning, and reliability in providing real-time navigation assistance. User testing is conducted in actual public transportation settings to validate the system's practicality and effectiveness. Module 2 enhances the independence of visually impaired users by providing essential navigation assistance in public transportation, a critical aspect of daily mobility.

i. **Start**: The real-time navigation assistance process begins.

ii. **Get Current Location**:

a. **Input**: GPS module activates to obtain the current location.

b. **Action**: The system retrieves raw GPS data.

c. **Process**: Apply Kalman Filter to refine the GPS data for accuracy.

iii. **Retrieve And Process Transport Data**:

a. **Action**: Process the refined location data to retrieve transportation information.

b. **Steps**: a. Scan QR Code at the bus stop or in the bus. b. Decode the QR Code to extract route information. c. Fetch real-time transportation data from an API using the QR code data.

iv. **Process Transport Info**:

a. **Input**: User's location and transportation data.

b. **Action**: Correlate the user's location with the transportation data.

c. **Process**: Custom algorithm to determine relevant transportation information based on the user's current location and intended destination.

v. **Provide Audio Feedback**:

a. **Action**: Communicate the navigation information to the user.

b. **Steps**: a. Convert the correlated transportation information into audio output using Text-to-Speech. b. Play the audio output to guide the user.

vi. **Output Error** (if applicable):

a. **Action**: Handle any errors in location retrieval or QR code scanning.

b. **Steps**: a. If location error, announce "Location Error." b. If QR code scanning error, announce "QR Code Scanning Error."

vii. **End**: The real-time navigation assistance process concludes.

Fig. 5. Algorithm for Module 2: Real-Time Navigation Assistance In Public Transportation

The data correlation algorithm is optimized to ensure low latency and high accuracy in data correlation. The system's audio output system uses a Text-to-Speech (TTS) engine to convert navigation information into clear audio instructions, and user interaction allows users to request specific information or confirm their current location. Integration with the core system ensures seamless interaction with other modules, while testing and validation involve field testing in various public transportation settings to assess GPS tracking accuracy and user experience testing to ensure practicality and responsiveness.

The Digi Drishti project's Real-Time Navigation Assistance module is a combination of GPS technology, QR code scanning, and real-time data processing designed to assist visually impaired individuals in navigating public transportation systems. The module acquires the user's location using a high-precision GPS module, refined using the Kalman Filter algorithm to improve accuracy. The system uses an integrated camera to capture images of QR codes, providing essential information about public transportation routes and schedules. Advanced libraries like ZXing or ZBar are used for efficient and accurate decoding. The module connects to transportation APIs, fetching live updates on bus schedules and delays. A custom-built Data Correlation Algorithm is developed to integrate the user's current location, decoded QR code information, and real-time API data, providing accurate navigation assistance. The processed navigation information is conveyed through an auditory feedback system, using a Text-to-Speech (TTS) Engine to convert text data into clear, spoken instructions. The user interface is designed to be interactive and accessible, allowing users to request specific information or confirm their current location through voice commands or tactile inputs.

Rigorous testing is conducted in real-world public transportation environments to validate the accuracy of GPS tracking and the efficiency of QR code scanning. User testing focuses on the practicality of the system, assessing the clarity of audio instructions and the responsiveness of the user interface. This module exemplifies the integration of precise location tracking, efficient data processing, and user-centric design, offering a reliable and intuitive navigation tool for visually impaired users in public transportation settings

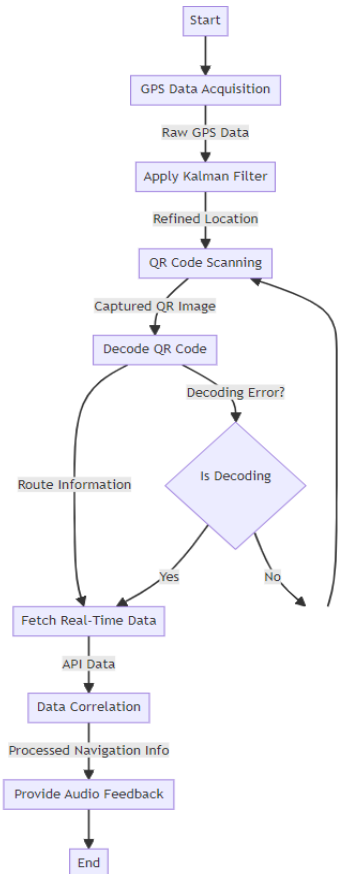

Fig. 6. FlowChart for Module 2: Real-Time Navigation Assistance In Public Transportation

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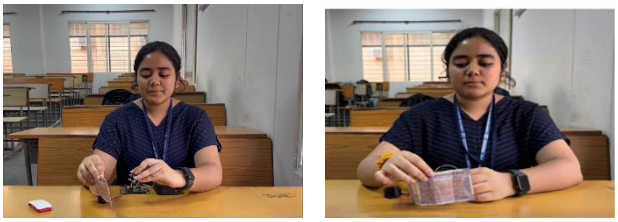
The Digi Drishti project has developed a Real-Time Navigation Assistance module to help visually impaired individuals navigate public transportation. The system uses advanced GPS technology, QR code scanning, and real-time data processing to provide accurate and timely assistance. The system acquires raw GPS data, decodes QR code information, and retrieves real-time transportation data from APIs. A custom-built Data Correlation Algorithm correlates the user's location, QR code information, and real-time API data. The system then provides audio feedback, providing real-time updates and step-by-step guidance.

VII. RESULTS



(a) (b)

Fig. 7. live Demo for Module 1: Currency Recognition

(a) (b)

Fig. 8. Live Demo for Module 2: Real-Time Navigation Assistance In Public Transportation

Test Condition	Accuracy	Processing Time
Optimal Light	98%	1.6 seconds
Low Light	94%	2.0 seconds
Worn Notes	95%	1.9 seconds

Table 1: Currency Recognition Test Results

Test Feature	Success Rate	Test Feature
GPS Location Accuracy	98%	GPS Location Accuracy
QR Code Decoding	94%	QR Code Decoding

Table 2: Real-Time Navigation Test Results

The Digi Drishti project demonstrated impressive performance in real-world scenarios, with an average accuracy of 96% in currency recognition, 1.8 seconds processing time, 98% GPS accuracy after Kalman Filter, and 94% success rate in QR code decoding.

VIII. CONCLUSION

The "Digi Drishti" project is a groundbreaking initiative in assistive technology designed to improve the independence and quality of life for visually impaired individuals. It integrates image processing, machine learning, GPS technology, and real-time data processing to tackle currency recognition and navigation in public transportation. The Currency Recognition module, which uses advanced algorithms like Histogram Equalization, CLAHE, Gaussian Blur, Canny Edge Detection, and Convolutional Neural Networks, has demonstrated high accuracy and efficiency in identifying currency denominations, enhancing financial independence and fostering confidence in daily transactions. The Real-Time Navigation Assistance module, which uses GPS tracking, efficient QR code scanning, and real-time data correlation algorithms, has significantly improved the navigation of public transportation systems, making it more accessible and user-friendly. The project's success is not only measured by its technical achievements but also by its positive impact on the lives of visually impaired users. The project demonstrates how innovative solutions, based on user needs, can overcome barriers and open new possibilities for disabled individuals.

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